

Scalability & Future Plan

Date	03 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application predicts HDI using only four input features.	Prediction accuracy may be limited because other socio-economic factors are not considered.	High
2	The application is currently deployed only on a local Flask server.	Users cannot access the system online from different locations.	High
3	The prediction model uses a Linear Regression algorithm, which may not capture complex relationships in the data.	Prediction performance may be lower than advanced machine learning models.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of local users through the Flask development server.	Deploy the application on a cloud platform (e.g., Render, Heroku, or AWS) to support multiple concurrent users.
2	Data Storage	Uses a static CSV dataset stored locally.	Integrate a relational database (e.g., MySQL or PostgreSQL) to store and manage updated HDI data.
3	Performance	Predictions are generated using a locally stored machine learning model.	Optimize the model and use a production-ready web server (e.g., Gunicorn or Waitress) for improved performance.
4	Security	No user authentication or secure communication is implemented.	Add user authentication, HTTPS, and input validation to enhance application security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase1	Developed the Human Development Index (HDI) Prediction System using a Linear Regression model and deployed it as a Flask web application with a user-friendly interface.	Completed (July 2026)	Provides users with an easy-to-use application for predicting the Human Development Index based on key socio-economic indicators.
Phase 2	Deploy the application on a cloud platform such as Render or AWS for online access.	Within 1 month	Enables remote access and supports multiple users.
Phase 3	Improve the prediction model by incorporating additional socio-economic indicators and evaluating advanced machine learning algorithms.	Within 2 months	Improves prediction accuracy and overall model performance.
Phase 4	Enhance the application with interactive visualizations, updated datasets, and improved user experience.	Within 3 months	Provides better insights, improved usability, and a more informative prediction system.